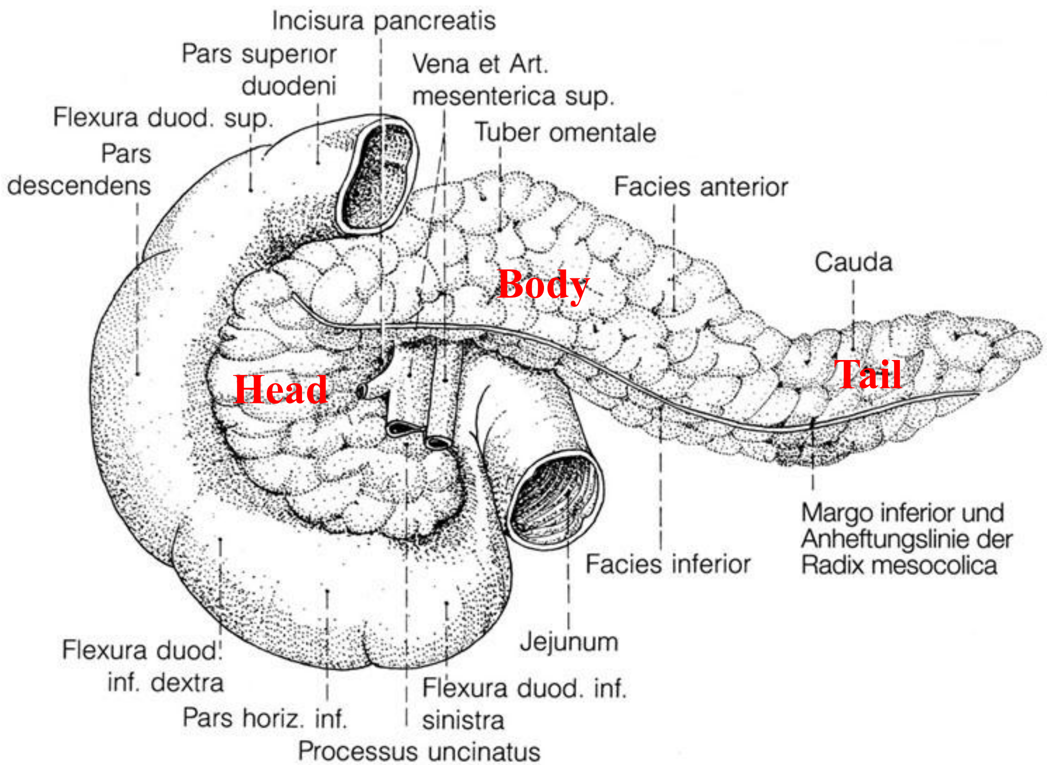
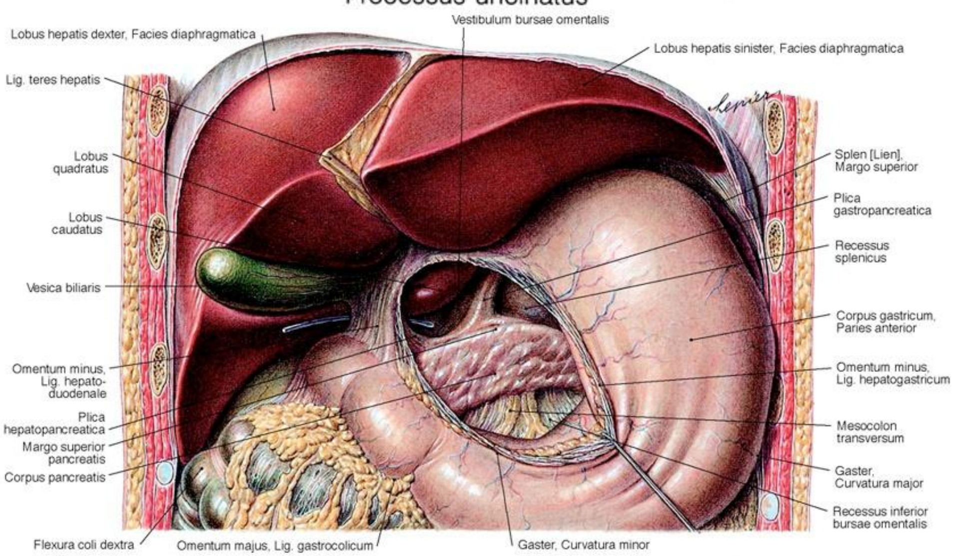
# Pancreas




*Functions:* **exocrine digestion** and **endocrine blood sugar regulation**.
It produces enzymes (amylase, lipase, protease) to break down food in the small intestine and secretes hormones, primarily insulin and glucagon, to manage glucose metabolism.

40-120 g, 13-15 cm

Head, body, tail

**Head (Caput pancreatis):**

4-8 cm wide

Lies in the duodenal loop

*Uncinate process*

*Pancreatic notch*

**Body (Corpus pancreatis):**

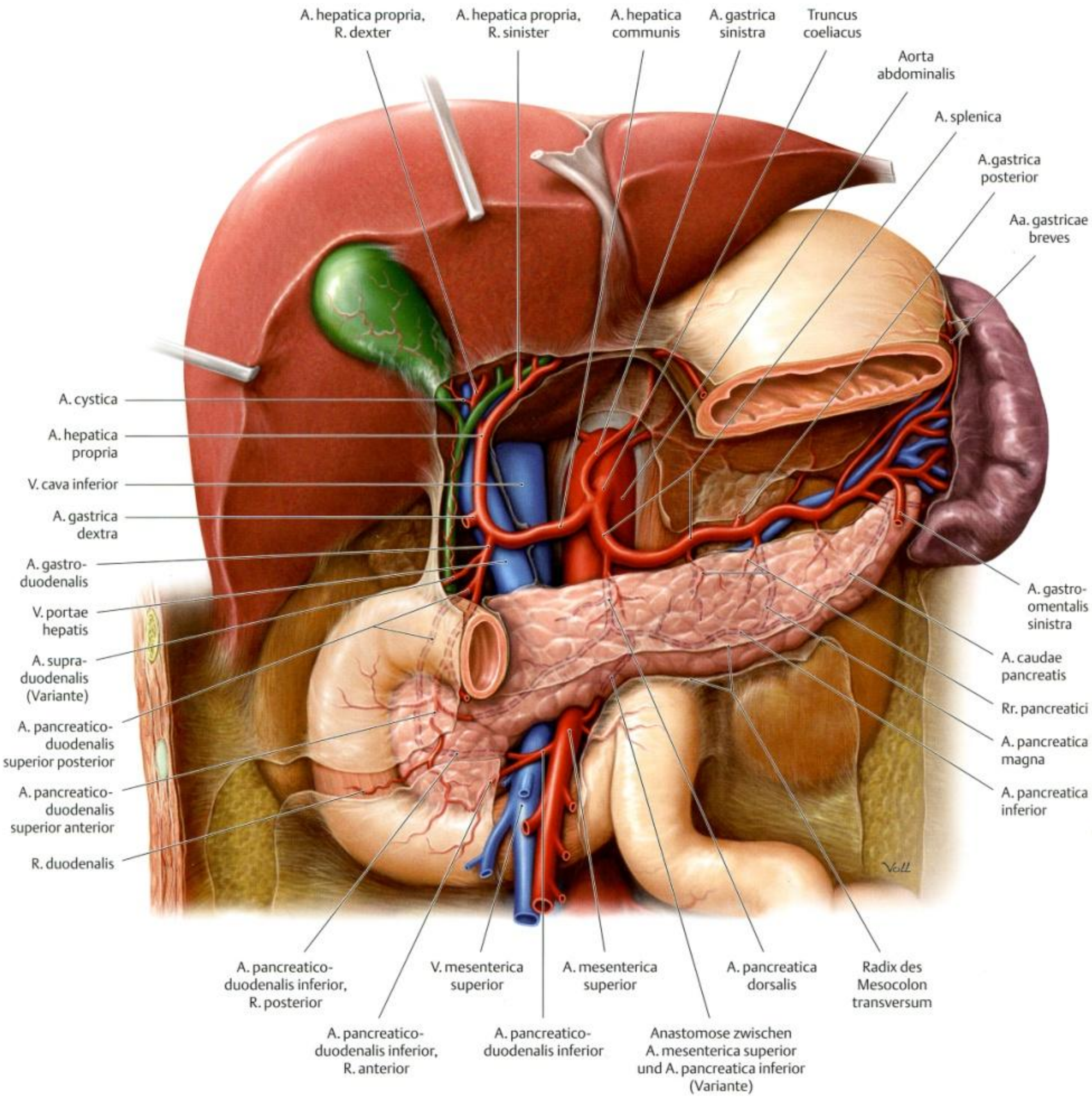
Facies post., ant., inf.

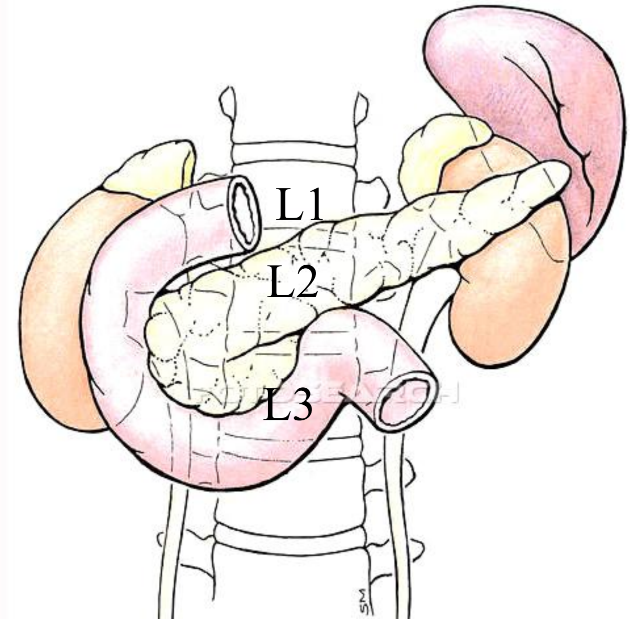
*Omental tuberosity*

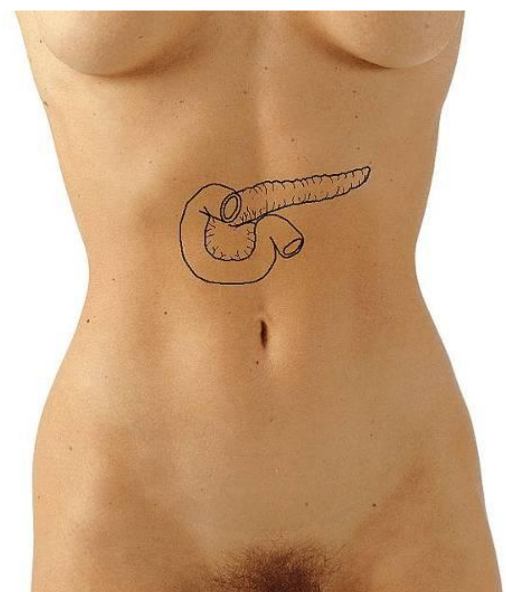
**Tail (Cauda pancreatis):**

It reaches the hilum of the spleen

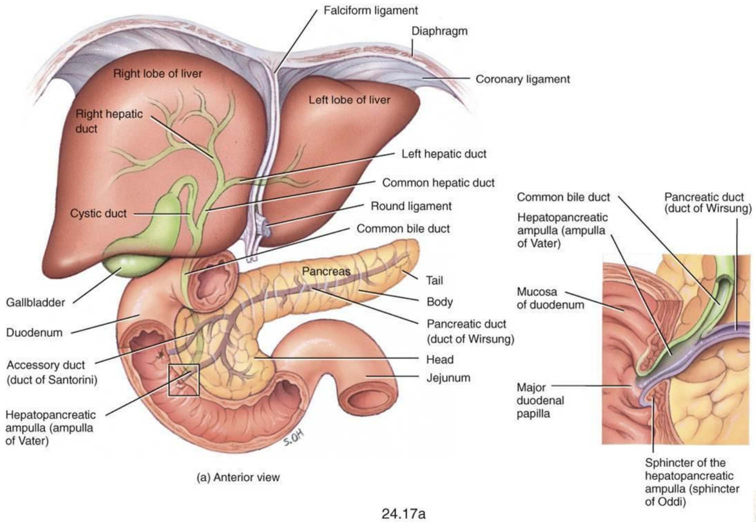
# Pancreas - vertebral levels

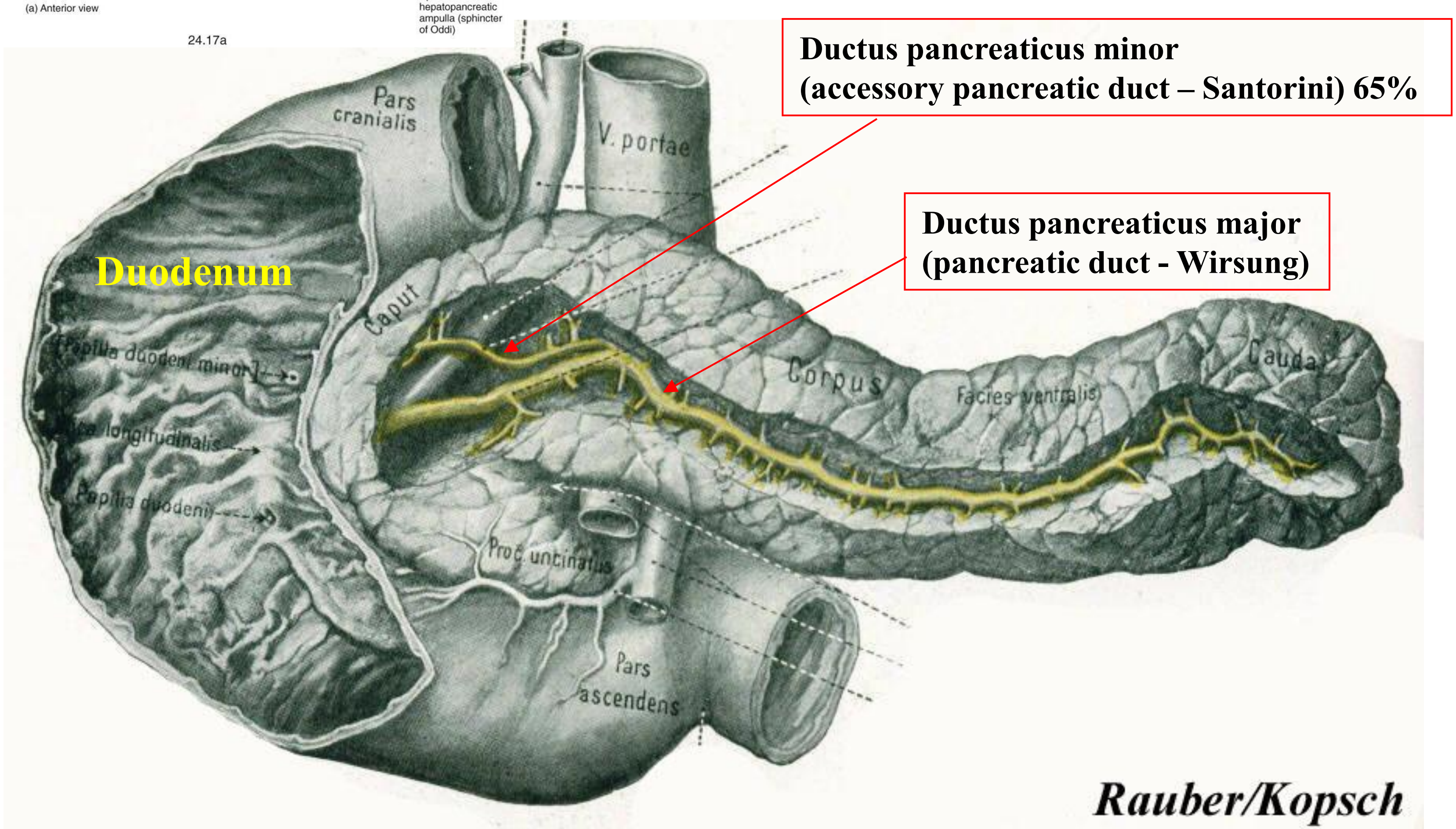


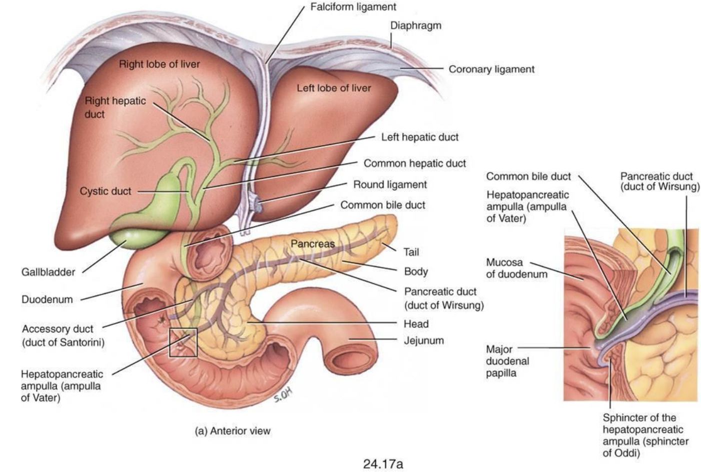



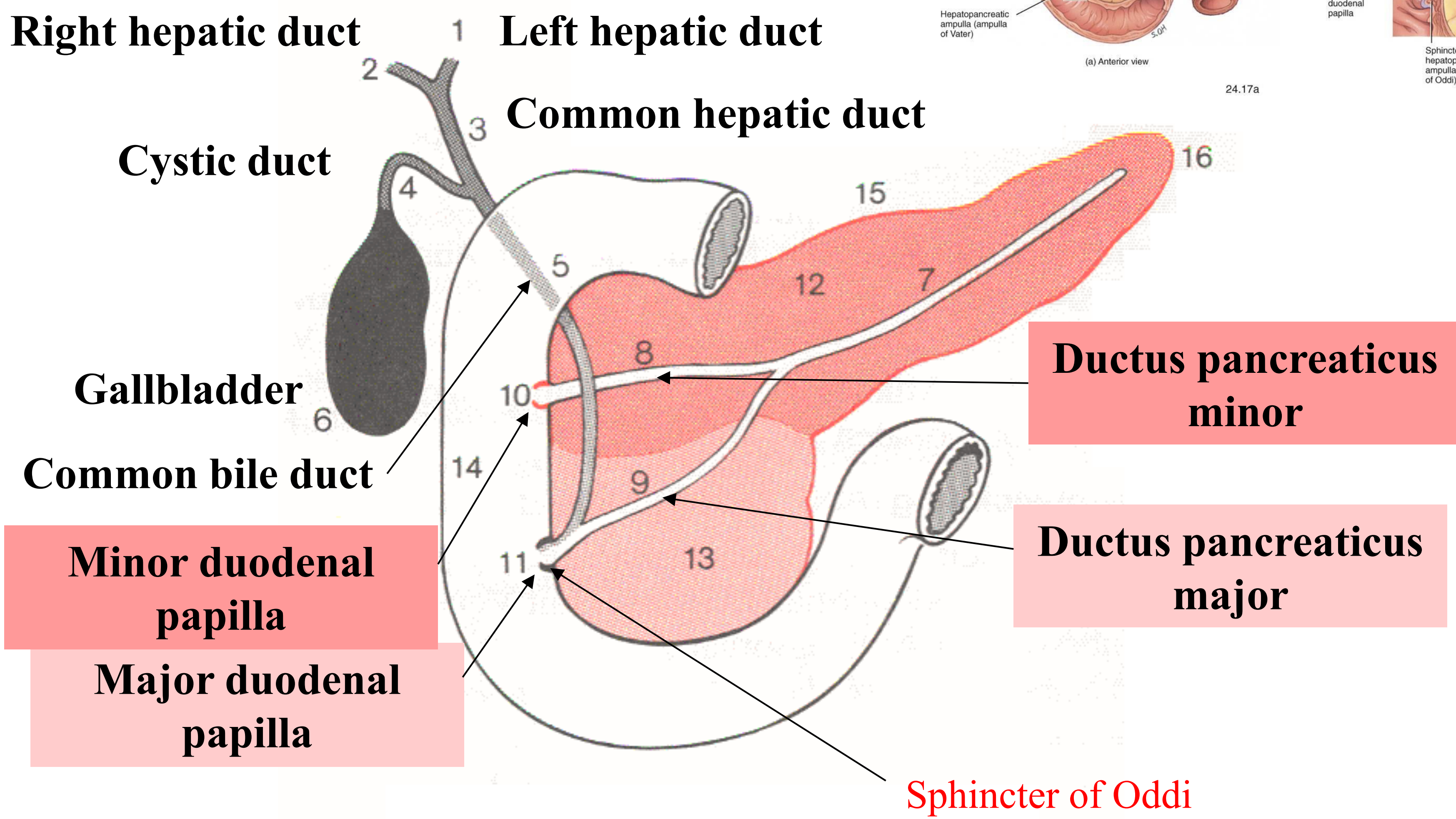
# Pancreas – ducts (exocrine part)



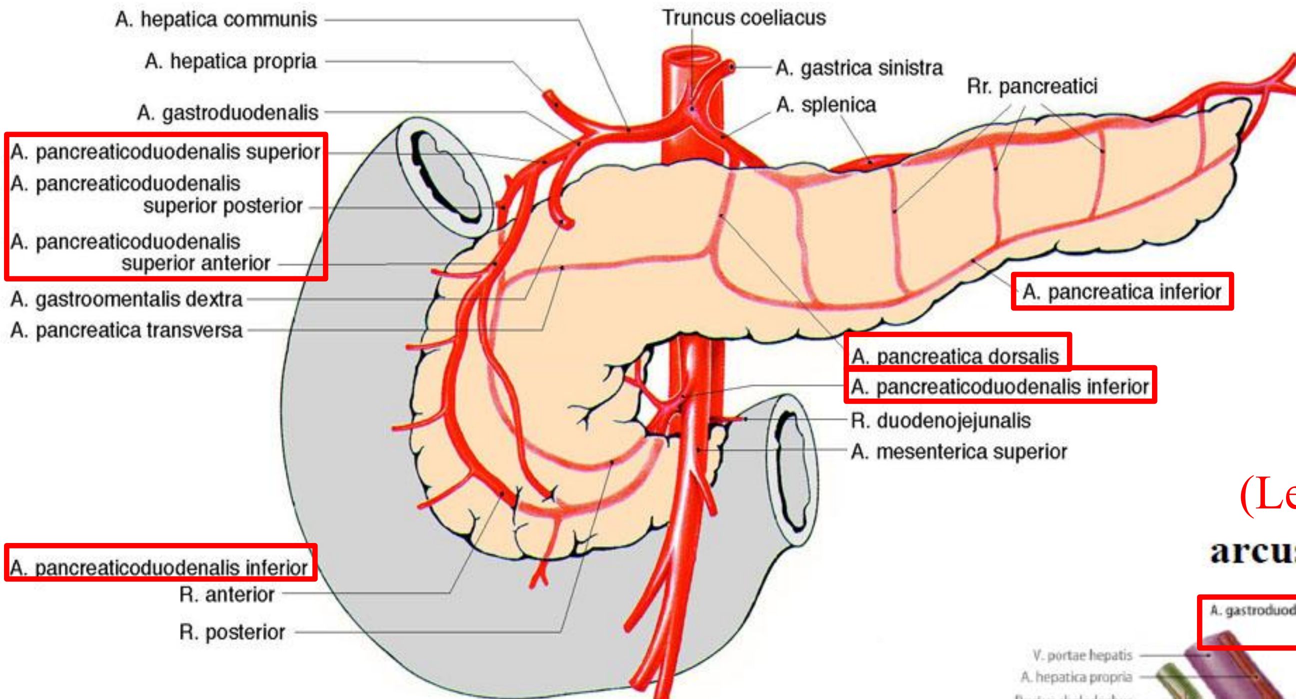



# Duodenal openings and the pancreatic ducts



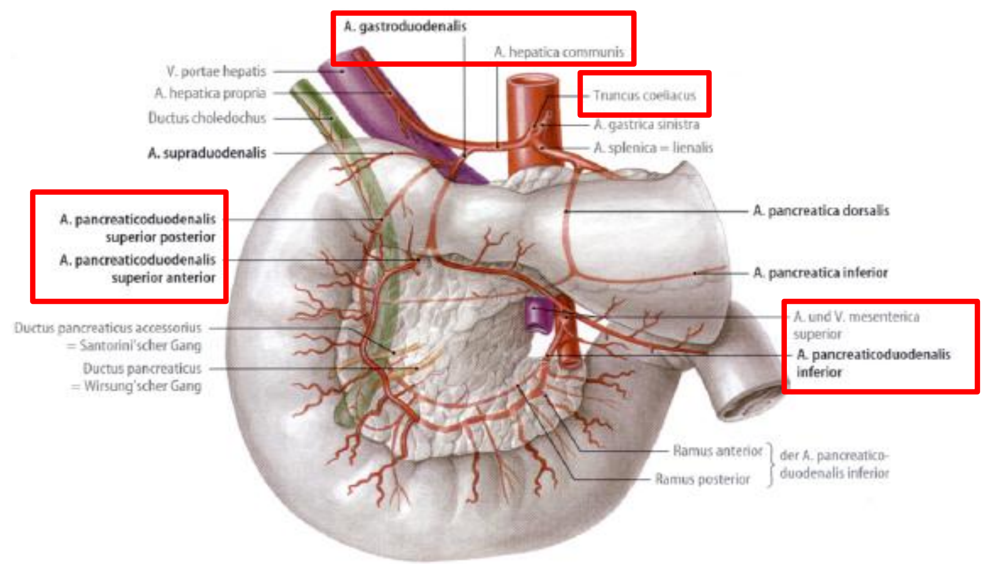



# Pancreas and duodenum – blood supply




(Lesser Riolan's anastomosis)

**arcus Riolani minor**

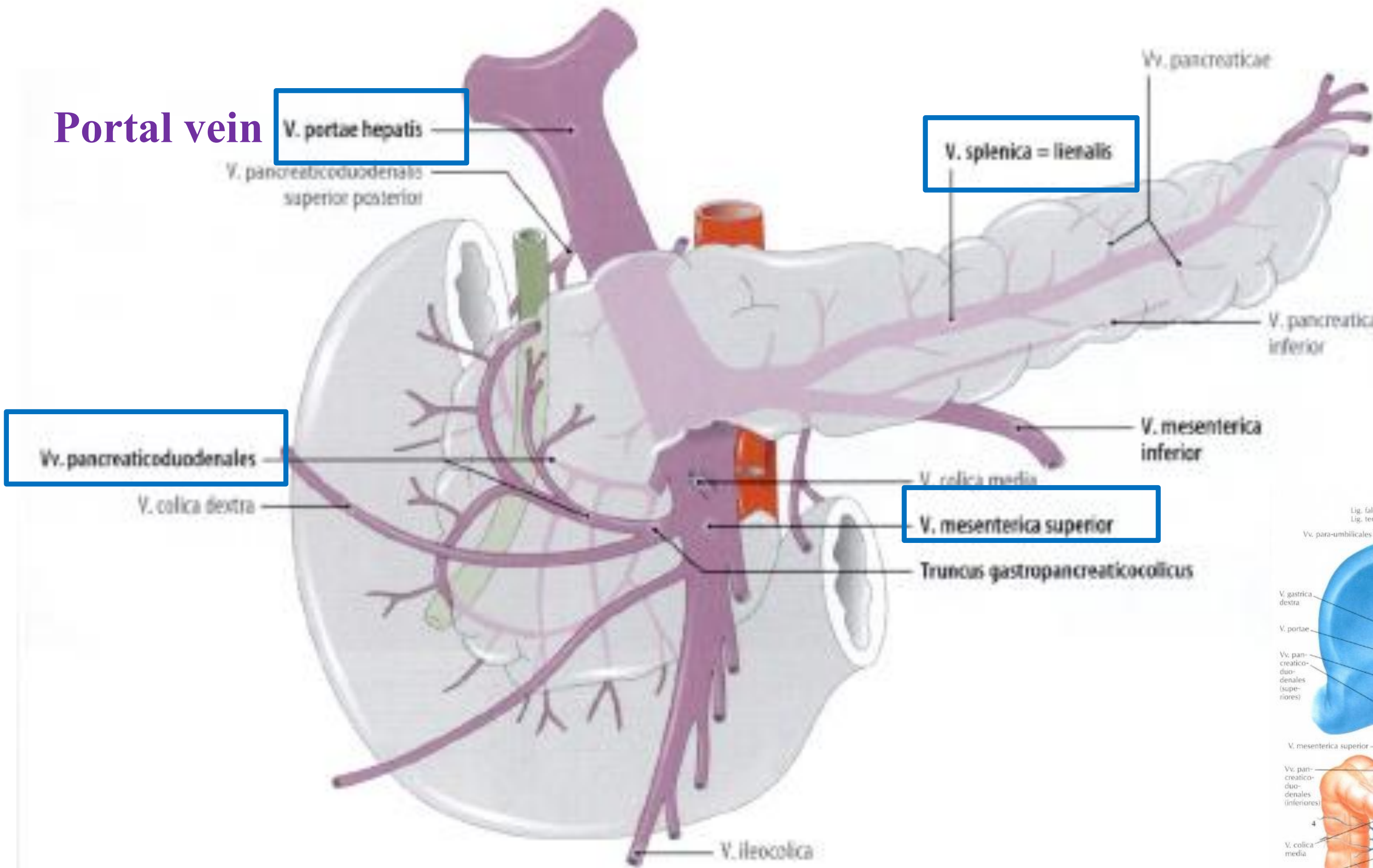



**Blood supply of the duodenum and the head of the pancreas**

Tillmann, Atlas der Anatomie, Springer 2005

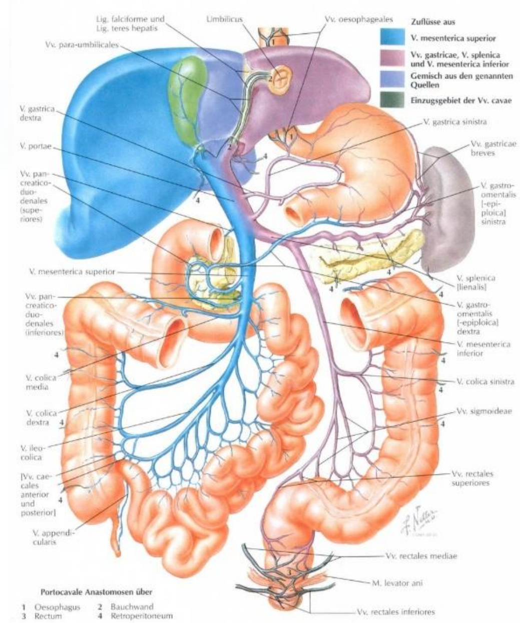
# Venous drainage of the duodenum and the pancreas

Portal vein

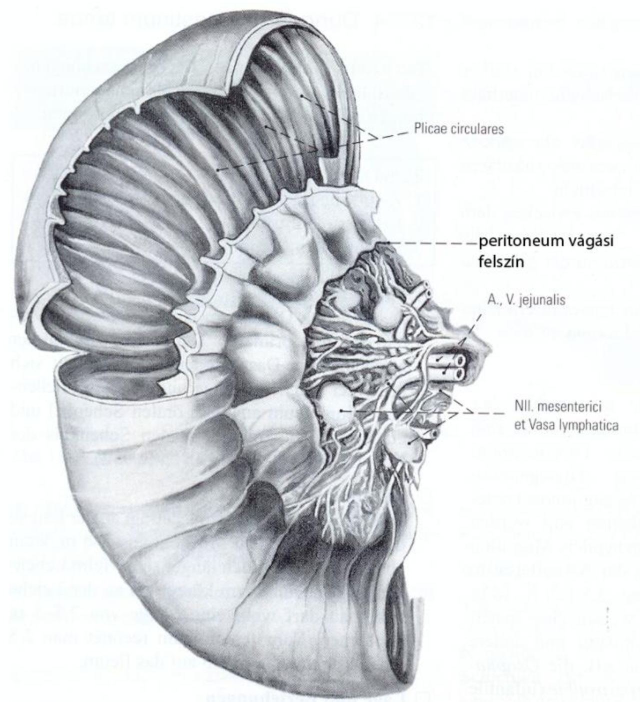


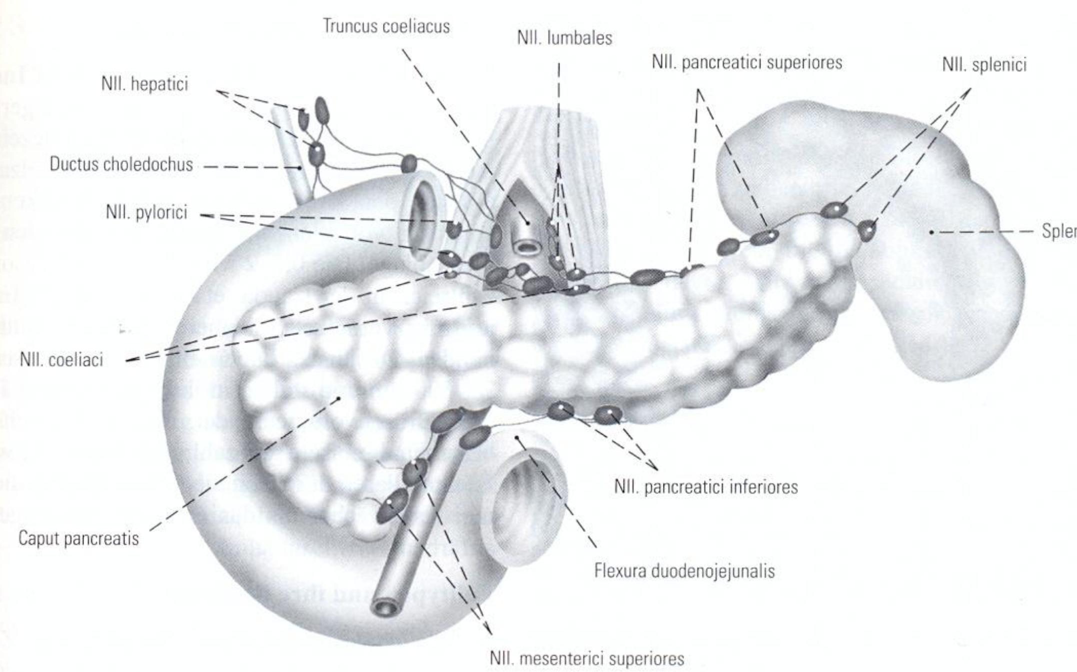

Tillmann, Atlas der Anatomie, Springer 2005

Portal vein



# Lymphatic drainage



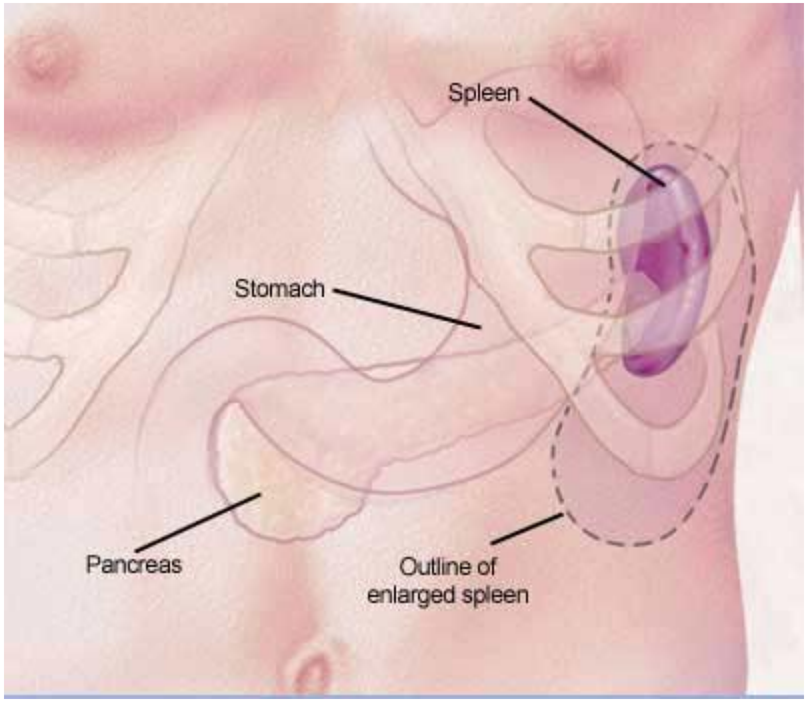

•**Head and Uncinate Process:** Lymphatic flow proceeds to ***superior mesenteric nodes***, ***anterior/posterior pancreaticoduodenal nodes, and pyloric nodes***, following the inferior pancreaticoduodenal vessels.

•**Body and Tail:** Lymph drains via the ***pancreaticosplenic nodes*** located along the splenic artery and vein to the ***celiac nodes.***

•**Final Common Routes:** The ultimate drainage destination for the entire pancreas includes the ***celiac, superior mesenteric, para-aortic, and aortocaval lymph nodes***.

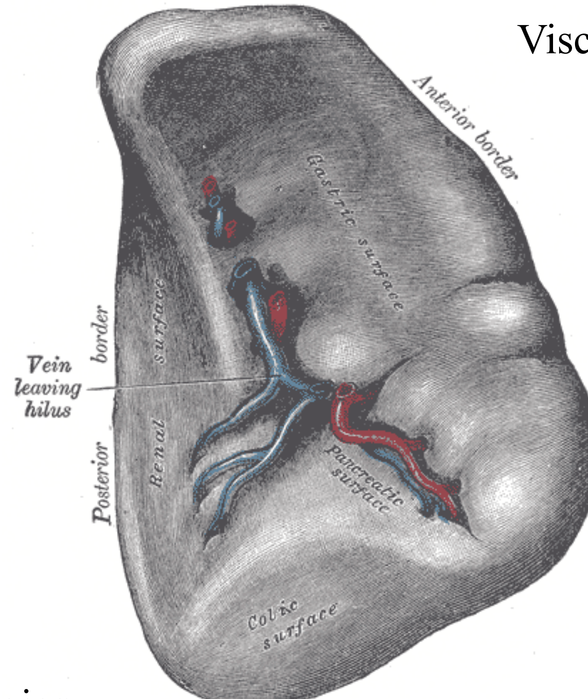
**Nomenclature and Lymphatic Drainage Patterns of Abdominal Lymph Nodes** Hyun Seok Cho [1], Jhii-Hyun Ahn PMCID: PMC9748447 PMID: 36545415

# Spleen – position, surfaces




Intraperitoneal organ in the left hypochondriac region

Between 9th and 11th ribs




Visceral surface:

-Renal
-Colic
-Gastric
-pancreatic

Diaphragmatic or costal surface

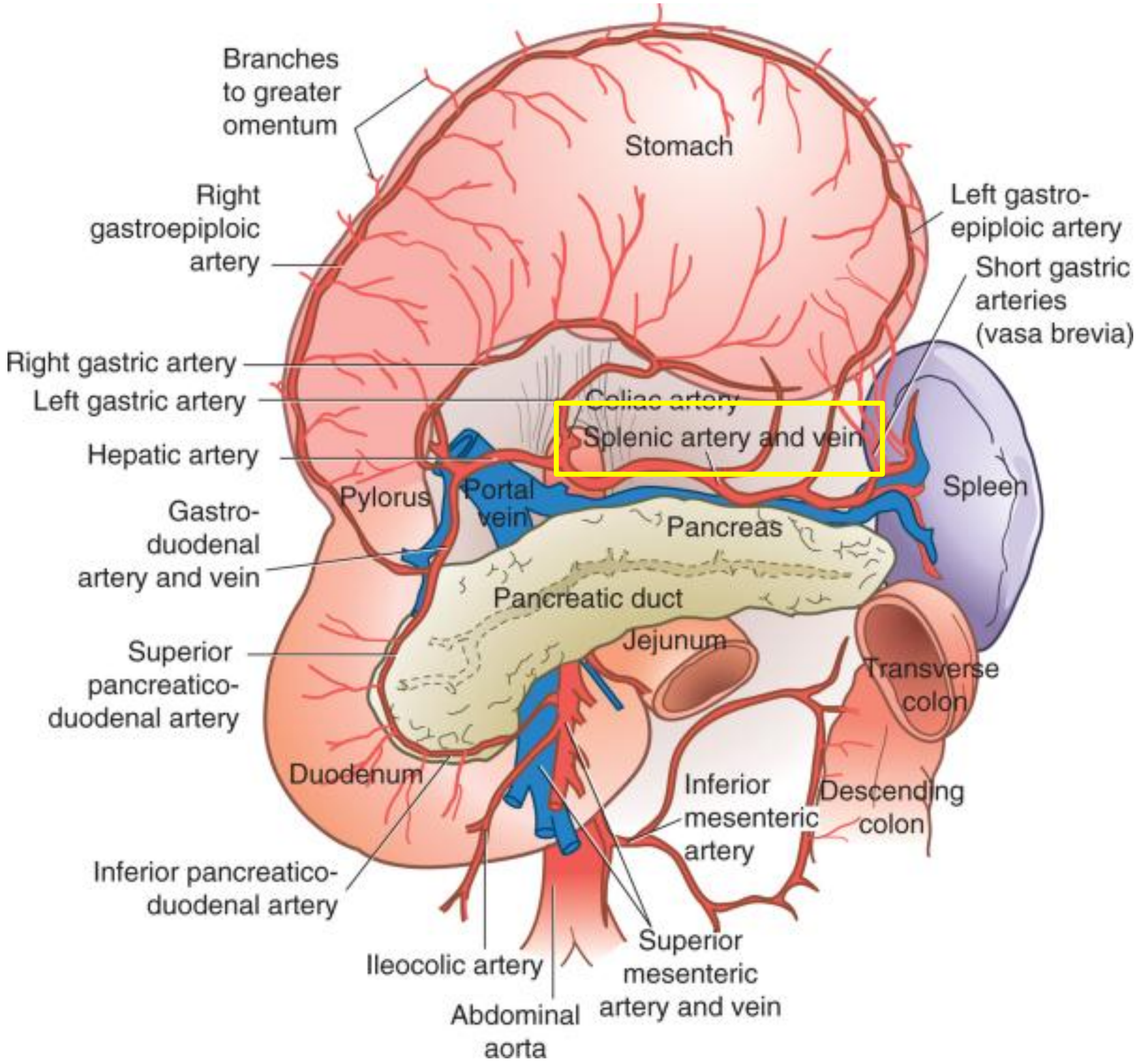
**Functions:**

**- blood filtration:** removing aged, damaged, or abnormal red blood cells from circulation.

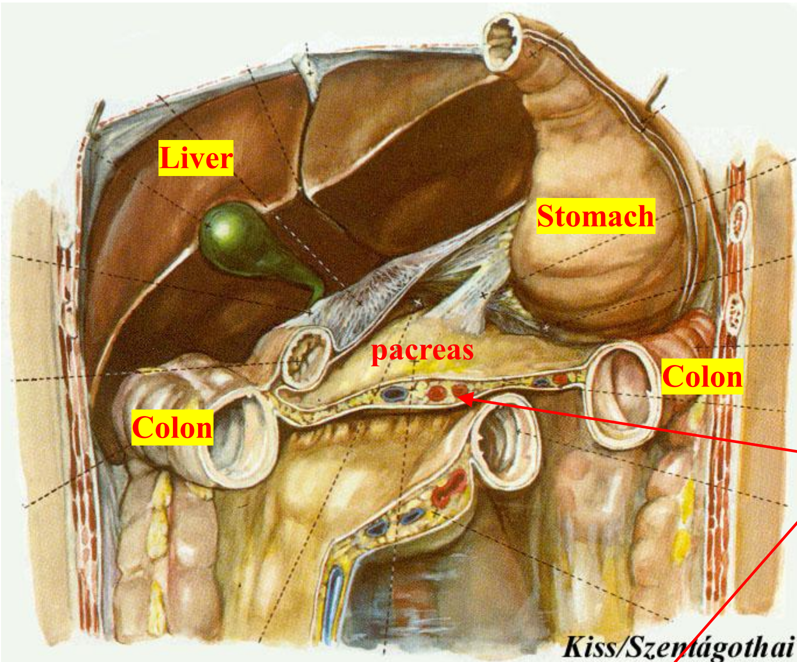
**- immune system support:** it is a key part of the lymphatic system.

**- iron recycling:** breaks down old red blood cells and recycles the iron, sending it back to the bone marrow for hemoglobin production.

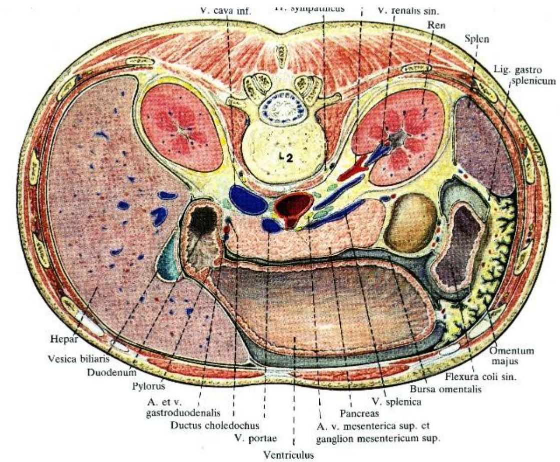
# Spleen - blood supply and venous drainage



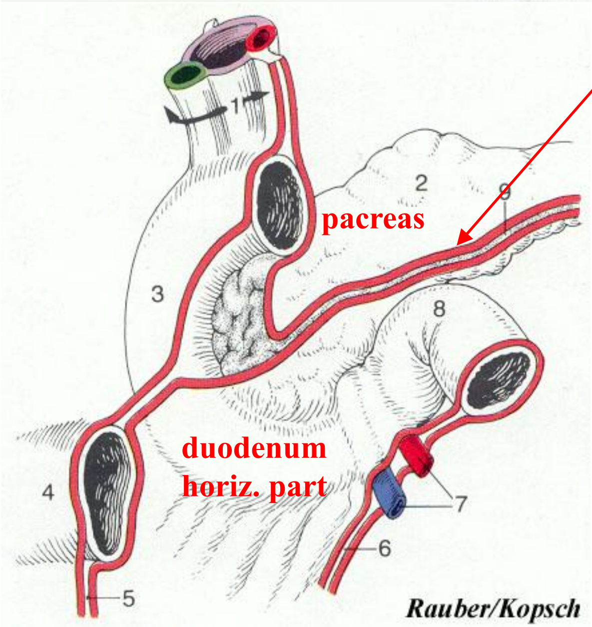
# Pancreas and duodenum – peritoneal position

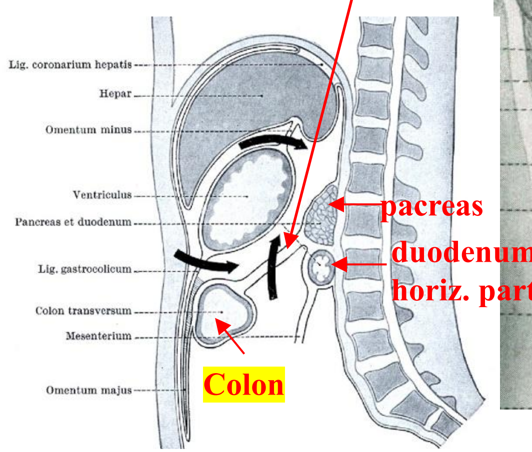



**Retroperitoneal**

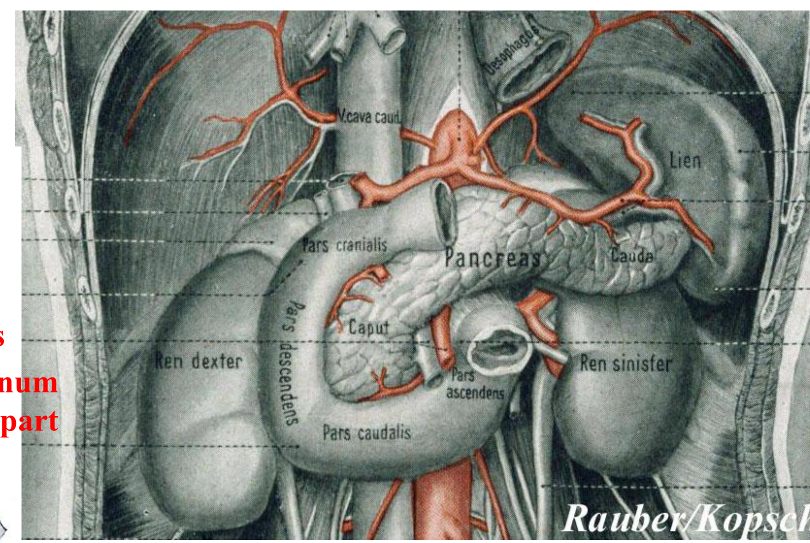



Transverse mesocolon



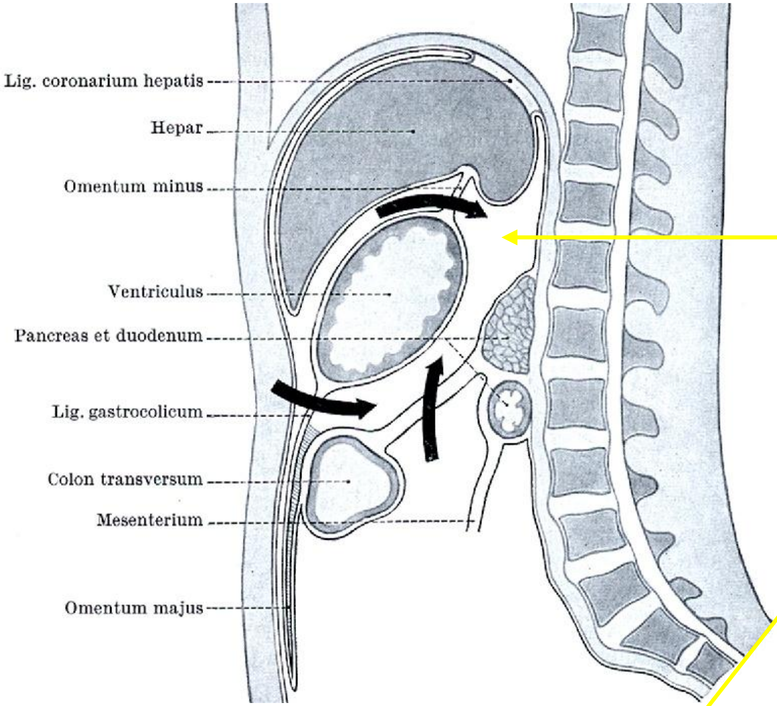




Midsagittal section



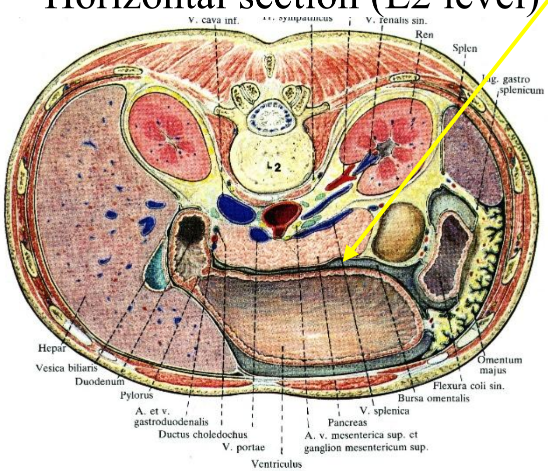
# Pancreas – peritoneal position

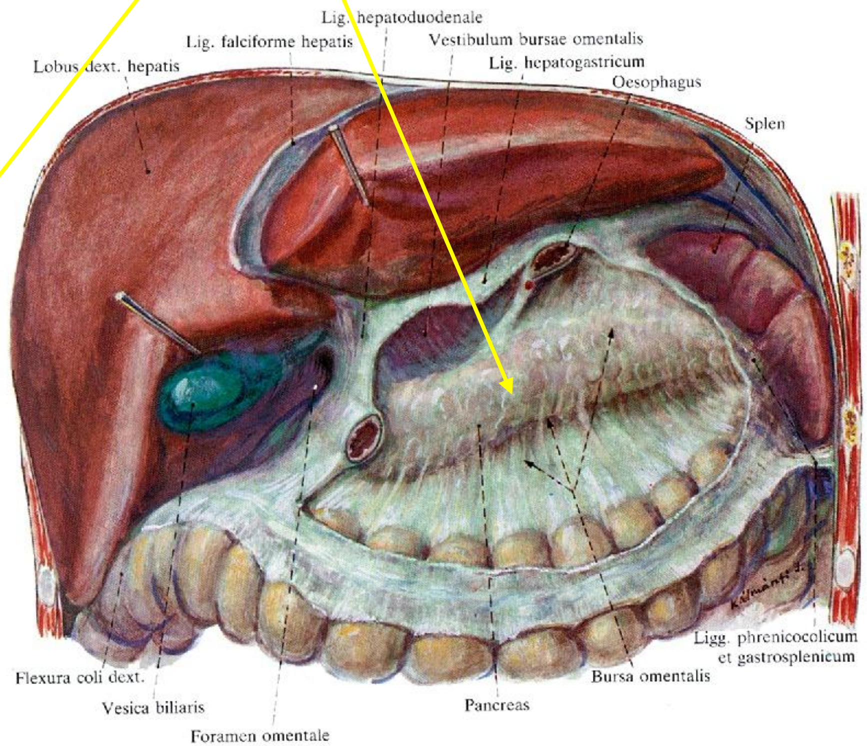
Midsagittal section




Behind the posterior wall of the omental bursa

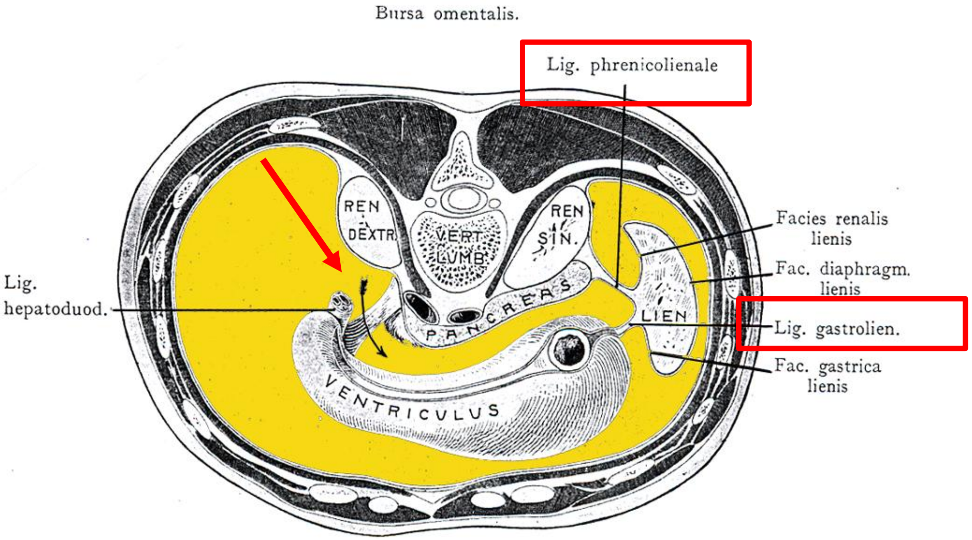
Horizontal section (L2 level)



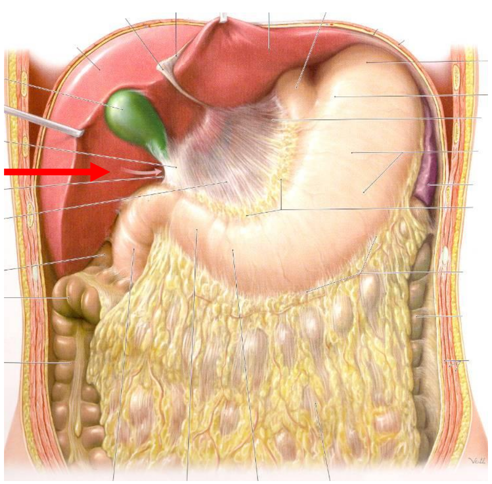



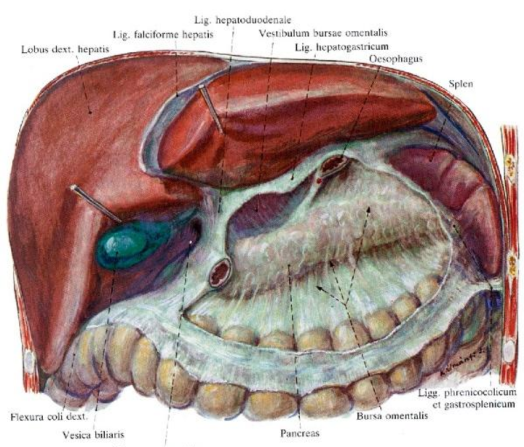
# Omental bursa

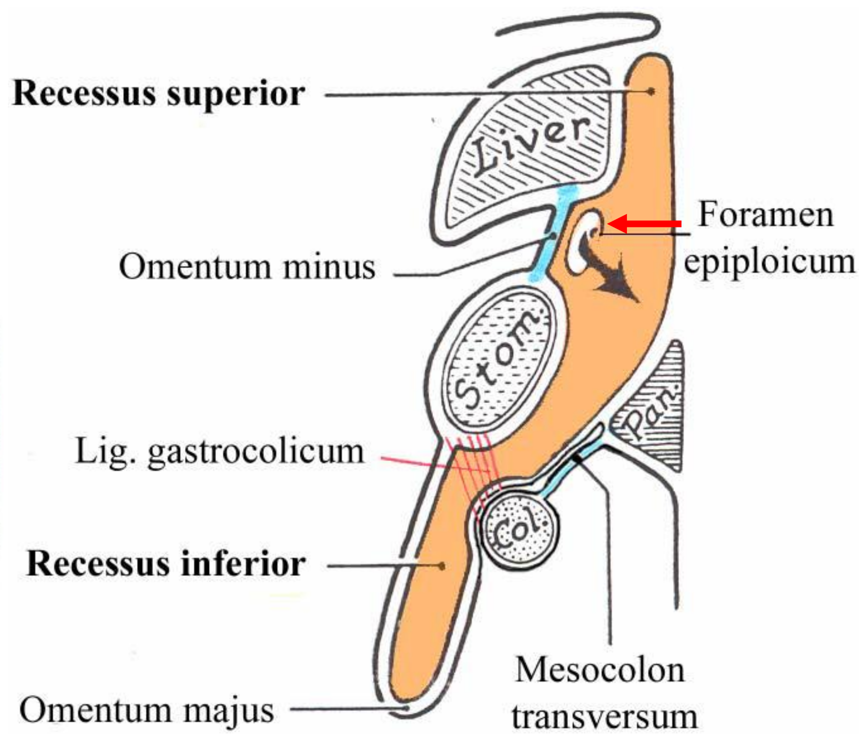
The omental bursa [or **lesser sac**], is a small, closed-off part of the peritoneal cavity, situated posterior to the stomach and lesser omentum, and anterior to the pancreas.




- **Epiploic foramen (of Winslow) !!!**
- **Vestibule**
- **Gastropancreatic fold**
- **Proper cavity**
- **Recesses: sup., inf. and splenic**





# Cross section through the L2 vertebra level